

Project – Final Group Report

Total Mark:	- 70 Marks (17.5% of the total course grade – assessed alongside the Group Presentation) - Learn@Seneca Submission (Due: Tuesday April 7 at 8:00AM)
Submission file(s):	- Project-GroupReport.docx (this document with your answers) - Project-Code (e.g., a public link to a shared folder, your website, or GitHub repository)

This milestone is your opportunity to present and showcase your completed project. Alongside your video presentation, each group must submit a written report that documents your project in detail. This report should clearly communicate your problem, approach, contributions, and results.

Group 7 - Visualizing Galactic Feature Extraction using Grad-CAM

Problem Definition

Our group set out to automate the morphological classification of galaxies, a task traditionally performed by human experts or large-scale citizen science projects like Galaxy Zoo. As modern sky surveys, such as those from the Vera C. Rubin Observatory, generate petabytes of data containing billions of celestial objects, manual inspection has become a significant bottleneck in the research pipeline. By developing a deep learning framework—transitioning from a ResNet18 baseline to a high-performance EfficientNet architecture—we aim to provide a scalable, high-accuracy solution for categorizing galaxies into Elliptical, Spiral, or Irregular classes. This is critical for modern astrophysics because a galaxy's shape is a direct record of its formation history, star-formation rate, and interaction with dark matter; automating this process allows astronomers to analyze cosmic evolution at a scale and speed previously impossible, turning raw visual data into actionable scientific insights. Our goal was originally to achieve a model accuracy of >70%, but that shifted to >95% after a successful baseline classifier.

Dataset

The data set chosen for this project was [Galaxy Zoo – The Galaxy Challenge](#) on Kaggle. The Galaxy Zoo website is a website where users classify galaxies from the Sloan Digital Sky Survey by means of a decision tree. The goal of The Galaxy Challenge was to create a model that accurately matches the vote distribution of classified galaxies. The dataset itself contains 61,578 labelled training images and 79,975 unlabelled testing images (labels only known to the competition hosts). The images themselves are RGB 424 x 424 JPEGs and the dataset included a CSV file that matched the image ID to its the vote distribution.

However, many factors including image quality and adjusting the ground truth to better match the functionality of Grad-CAM, meant that the total number of images used in this project was 20,529. A 80/10/10 split was used for train/validation/test, making the final training set size 16,423, and the final validation and test size to be 2,053.

Ground Truth

Several challenges were presented with regards to the dataset's labels. Firstly, the format of vote distributions were not readable. They were presented as the percentage of responses for each leaf of each node. It was unclear whether an image was classified as Spiral, Elliptical, or Irregular from the data. Secondly, Grad-CAM functions best when there are hard labels. Therefore, it was important to translate the numeric labels into hard classes. But, this introduces error since it can distort the original ground truth of the dataset (ie. A galaxy is given the label of "Elliptical" for this project, but its original label more closely aligns with "Irregular"). Ergo, it was important to find a method that maintained the integrity of the original labels.

	GalaxyID	Class1.1	Class1.2	Class1.3	Class2.1	Class2.2	Class3.1	\
0	100008	0.383147	0.616853	0.000000	0.000000	0.616853	0.038452	
1	100023	0.327001	0.663777	0.009222	0.031178	0.632599	0.467370	
2	100053	0.765717	0.177352	0.056931	0.000000	0.177352	0.000000	
3	100078	0.693377	0.238564	0.068059	0.000000	0.238564	0.109493	
4	100090	0.933839	0.000000	0.066161	0.000000	0.000000	0.000000	
	Class3.2	Class4.1	Class4.2	...	Class9.3	Class10.1	Class10.2	\
0	0.578401	0.418398	0.198455	...	0.000000	0.279952	0.138445	
1	0.165229	0.591328	0.041271	...	0.018764	0.000000	0.131378	
2	0.177352	0.000000	0.177352	...	0.000000	0.000000	0.000000	
3	0.129071	0.189098	0.049466	...	0.000000	0.094549	0.000000	
4	0.000000	0.000000	0.000000	...	0.000000	0.000000	0.000000	
	Class10.3	Class11.1	Class11.2	Class11.3	Class11.4	Class11.5	Class11.6	
0	0.000000	0.000000	0.092886	0.0	0.0	0.0	0.325512	
1	0.459950	0.000000	0.591328	0.0	0.0	0.0	0.000000	
2	0.000000	0.000000	0.000000	0.0	0.0	0.0	0.000000	
3	0.094549	0.189098	0.000000	0.0	0.0	0.0	0.000000	
4	0.000000	0.000000	0.000000	0.0	0.0	0.0	0.000000	
[5 rows x 38 columns]								

Sample of labels from training_solutions_rev1.csv from Galaxy Zoo – The Galaxy Competition on Kaggle

Willett et al. (2013) describes how redshift can distort apparent galaxy morphology and image quality and subsequently explains a method at flagging the highest quality examples of a class. This is the threshold method, in which a “clean” sample has vote fractions that exceed a given threshold showing that the question is well answered by the community (Willett et al., 2013). By cross referencing the decision tree provided in Willett et al. (2013) and the definition of the galaxy classes from Sparke & Gallagher (2007), 5 key leaves for class identification were found. They were:

Elliptical: 1.1 *“Is the galaxy simple smooth and rounded, with no sign of a disk? - Smooth”*

Spiral: 1.2 *“Is the galaxy simple smooth and rounded, with no sign of a disk? - Features or Disk”*, 4.1 *“Is there any sign of a spiral pattern? - Yes”*

Irregular: 1.3 *“Is the galaxy simple smooth and rounded, with no sign of a disk? - Star or Artifact”*, 6.1 *“Is there anything odd? - Yes”*

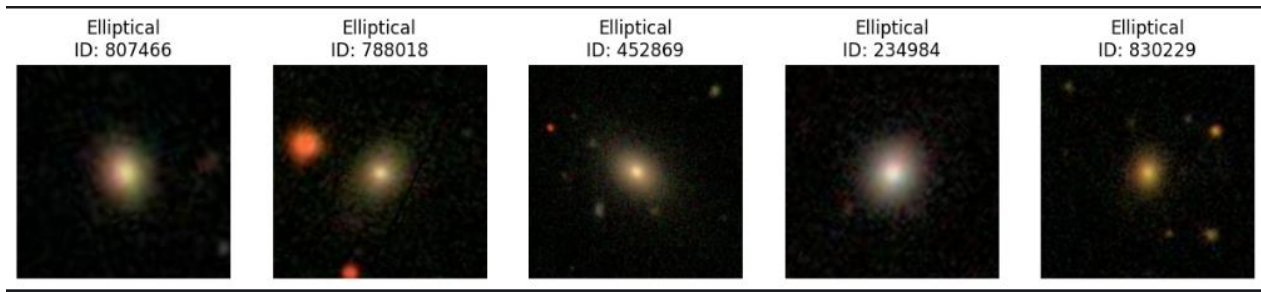
Where 1.1, 1.2, ect., is notation from Galaxy Zoo and Willett et al. (2013) depicting the question and response on the decision tree (Ex. 6.1 is sixth question, first response). From these leaves, 3 logic rules were made to assign classes to images:

$$\text{if } x_{1.1} \geq \theta_0 = \text{Elliptical}$$

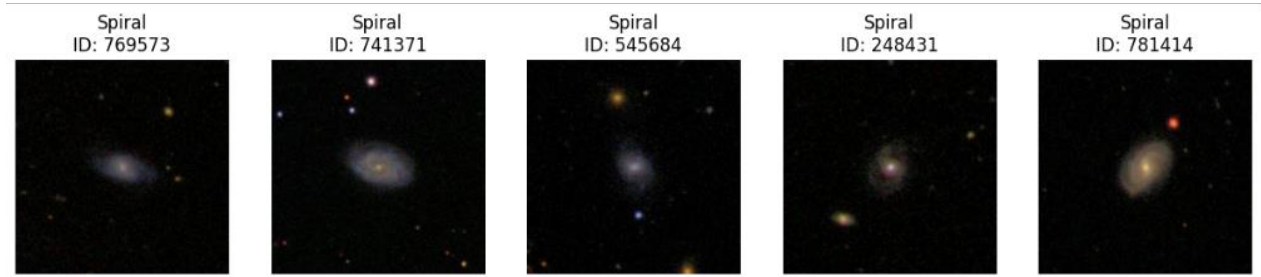
$$\text{if } x_{1.2} \geq \theta_0 \text{ and } x_{4.1} \geq \frac{5}{8}\theta_0 = \text{Spiral}$$

$$\text{if } x_{1.3} \geq \theta_0 \text{ or } x_{6.1} \geq \frac{7}{8}\theta_0 = \text{Irregular}$$

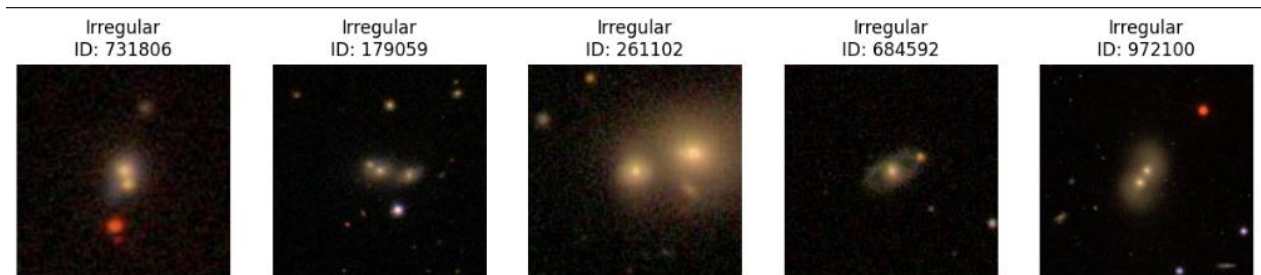
Where x is the value of the vote fraction of that leaf for the image and θ_0 is the threshold, 0.8 in this project. 5/8 and 7/8 are arbitrary constants designed to represent the split in total votes as one progress in the decision tree. Since less people are answering later questions, there is more variability in their vote fractions. Leaf 6.1 is high because it is more likely that the images that are unanimously odd are classified as irregular. The algorithm labelled the images as follows:



Elliptical Galaxies as labelled by the Threshold Algorithm



Spiral Galaxies as labelled by the Threshold Algorithm



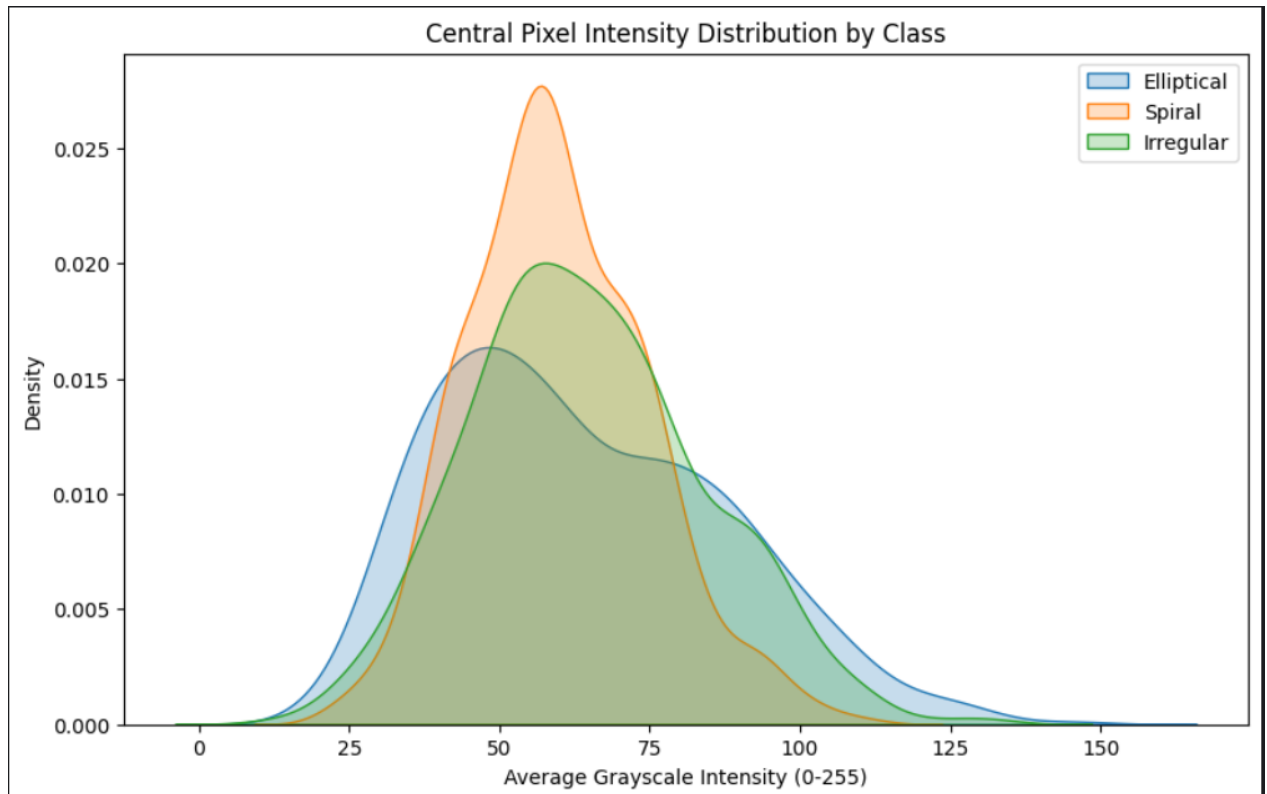
Irregular Galaxies as labelled by the Threshold Algorithm

While elliptical and spiral appear to be accurate to the original ground truth, there appears to some errors when it comes to the irregular class. For example, image 684592 in the above figure, which appears to have spiral features but was classified as Irregular. This deviation from the ground truth is a source of bias and can hinder the model's ability to learn the irregular class.

After assigning hard labels to the data, it was found that there was a class imbalance. 9107 spiral images, 8132 elliptical images, and 3290 irregular images.

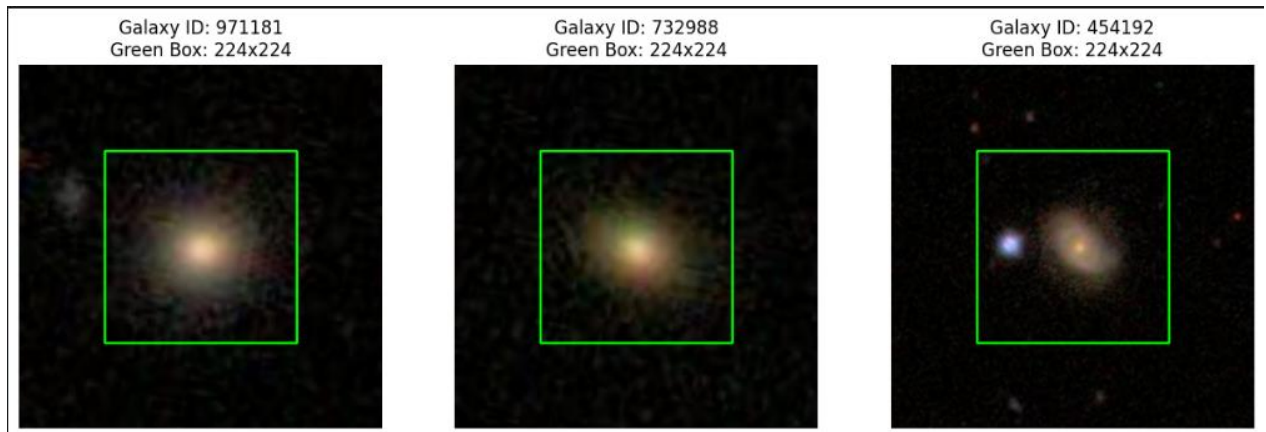
Dataset Splitting

A stratified split was generated to handle class imbalance. The image IDs were saved to CSV files to reproduce the sets between notebooks and prevent leakage. The images were kept in RGB due to colour related class features such as elliptical galaxies having older, redder stars. The histogram equalization of the data was analyzed and it was found that images were already equalized.



Central Pixel Intensity Distribution by Class

To prepare the data for the models, the images were normalized to the mean and standard deviation of the test set. Images were cropped to 224 x 224 pixels, which is standard for ResNet18 and EfficientNet. The crops were centered since all the images had the galaxies centered and to prevent valuable features from being removed in the crop. However, EfficientNet training had a RandomResizedCrop with a scale of (0.8, 1.0) and a ratio of (0.9, 1.1) to add some variability and hopefully improve performance.



Test Images with green boarder showing 224 x 224 center crop.

Since space has not direction, galaxies can follow any orientation. Therefore, RandomRotation, RandomHorizontalFlip, and RandomVerticalFlip were applied to the data. Finally, to handle class imbalance, weighted augmentation was added to the training set.

Previous Work

The evolution of automated galaxy classification has transitioned from the foundational morphological frameworks established by Sparke & Gallagher (2007) to modern crowdsourced and deep learning paradigms. Such as the Galaxy Zoo project, as detailed by Willett et al. (2013) and Banerji et al. (2010), which utilized community sourced classification. However, it became apparent that machine learning could greatly streamline the task of galaxy classification. This project builds directly on directly upon the Galaxy Zoo 2 dataset and the architectural challenges presented in the 2013 Kaggle Galaxy Challenge and on the comparative logic suggested by Nerella et al. (2026). This project focuses on a streamlined transition from a ResNet18 baseline to a more parameter-efficient EfficientNet-B0, incorporating Grad-CAM to provide the visual interpretability that early automated models lacked.

Method and Contributions

This project used Kaggle GPU T4x2 to accelerate training. Cross Entropy was the loss function. The objective was to first, complete exploratory data analysis and

preprocessing. Then train a baseline ResNet18 classifier model. After, train an EfficientNet-b0 classifier and spend time to improve the performance on the data. Finally, add Grad-CAM to the model to output model vision heatmaps. The roles were Tatum did the EDA/Preprocessing and ResNet18 training, Stephane did the EfficientNet-b0 training, and JohnPaul did the Grad-CAM implementation.

The ResNet18 model was not trained long since it was only to establish a benchmark. There were very few iterations and no hyperparameter tuning. The only trouble was when training crashed due to memory shortage, which was resolved by reducing batch size and the number of workers. EfficientNet was different. Since the goal was to improve performance, training was extensive and hyperparameters were frequently adjusted. Hyperparameters for ResNet18 and the best EfficientNet-b0 model are as follows:

ResNet18		EfficientNet-b0	
Image Size	224	Image Size	224
Batch Size	8	Batch Size	16
Epochs	8	Epochs	20
LR	1e-4	LR	1e-4
Weight Decay	1e-5	Weight Decay	1e-4
Number of Workers	0	Number of Workers	0

The process to complete Grad-CAM were as follows: First, Implement the Grad-CAM class. Within it, self initialization is done and hooks are registered, and functions for saving the activation, saving the gradient, generating the heatmaps, and deleting the hooks are defined. Next, another function is implemented to overlay the heatmap on to its respective image. When it comes time to actually utilize Grad-CAM, how we set it up depends on which model we're using; ResNet or EfficientNet. Finally, Generate the heatmap, prepare

the original image for display by undoing the normalization done in the heatmap generating step, and visualize the Grad-CAM.

The project did not emanate any previous work except for the Threshold Algorithm from Willett et al. (2013) as mentioned in “Ground Truth”.

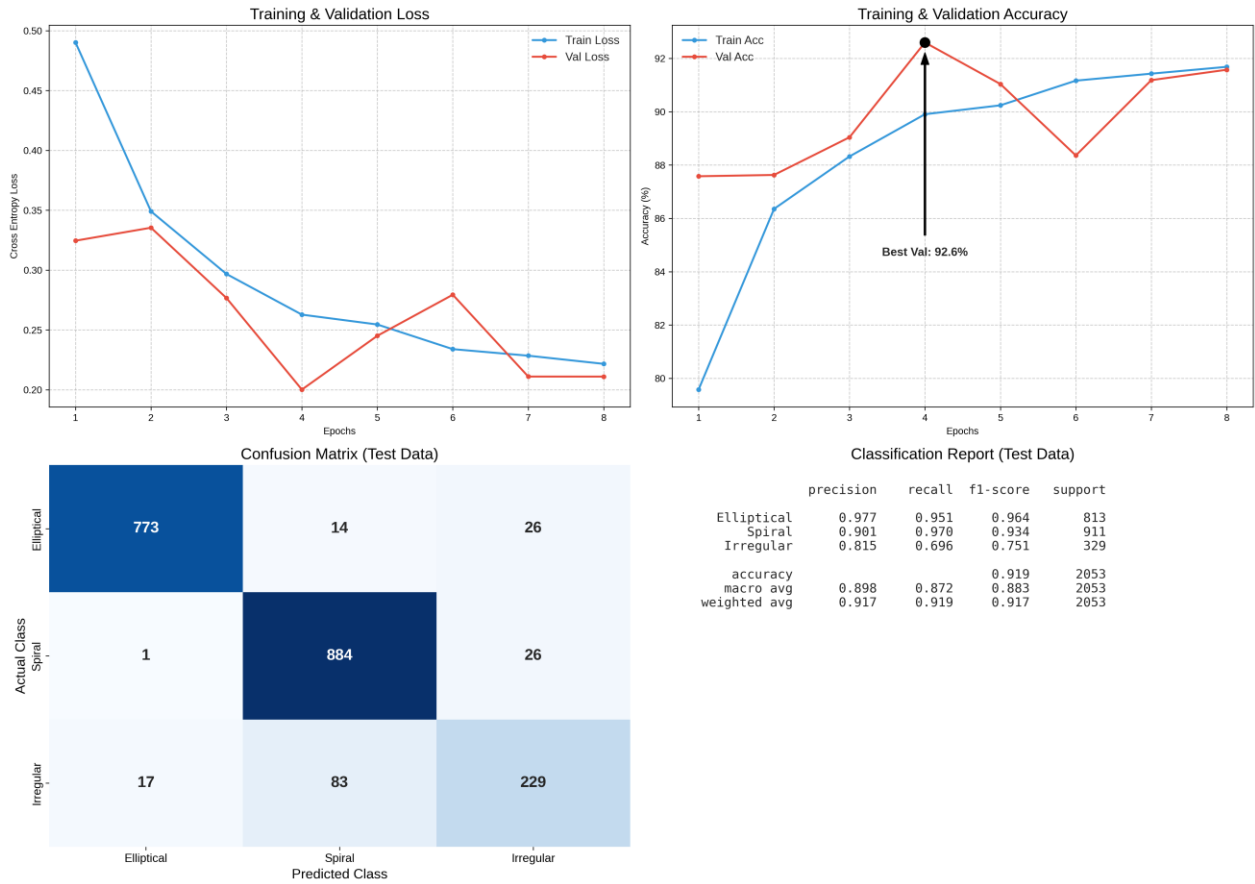
Outcome and Reflection

It is a victory that the model that works better than 70%, the initial goal. However, it is disappointing that the new benchmark of >95% could not be achieved. It is surprising that EfficientNet did not outperform ResNet18. EfficientNet, overall, took longer to train and it was more finicky. (Epochs took longer, the model took longer to converge) EfficientNet gives a smaller PTH file than ResNet which is because EfficientNet has less parameters than ResNet. Grad-CAM, while in general was difficult to implement, functioned without error when deployed.

One of the challenges faced, was the model’s poorly handling of the irregular class. This is possibly due to the bias from assigning the hard labels or from the fact that the irregular is the minority class and models tend to struggle with classifying minority classes. The best way to address this is to train the model on a different dataset with hard classes and equal class distribution. This was not an option because this group had progressed too far into the project to find a new dataset. Another option is to lower the threshold and retrain the model, this should hypothetically introduce more fringe cases and allow the model to better learn the boundaries between classes. Finally, this project could be about predicting a galaxy’s vote distribution instead of assigning a class and risk a more difficult Grad-CAM integration process.

Evaluation

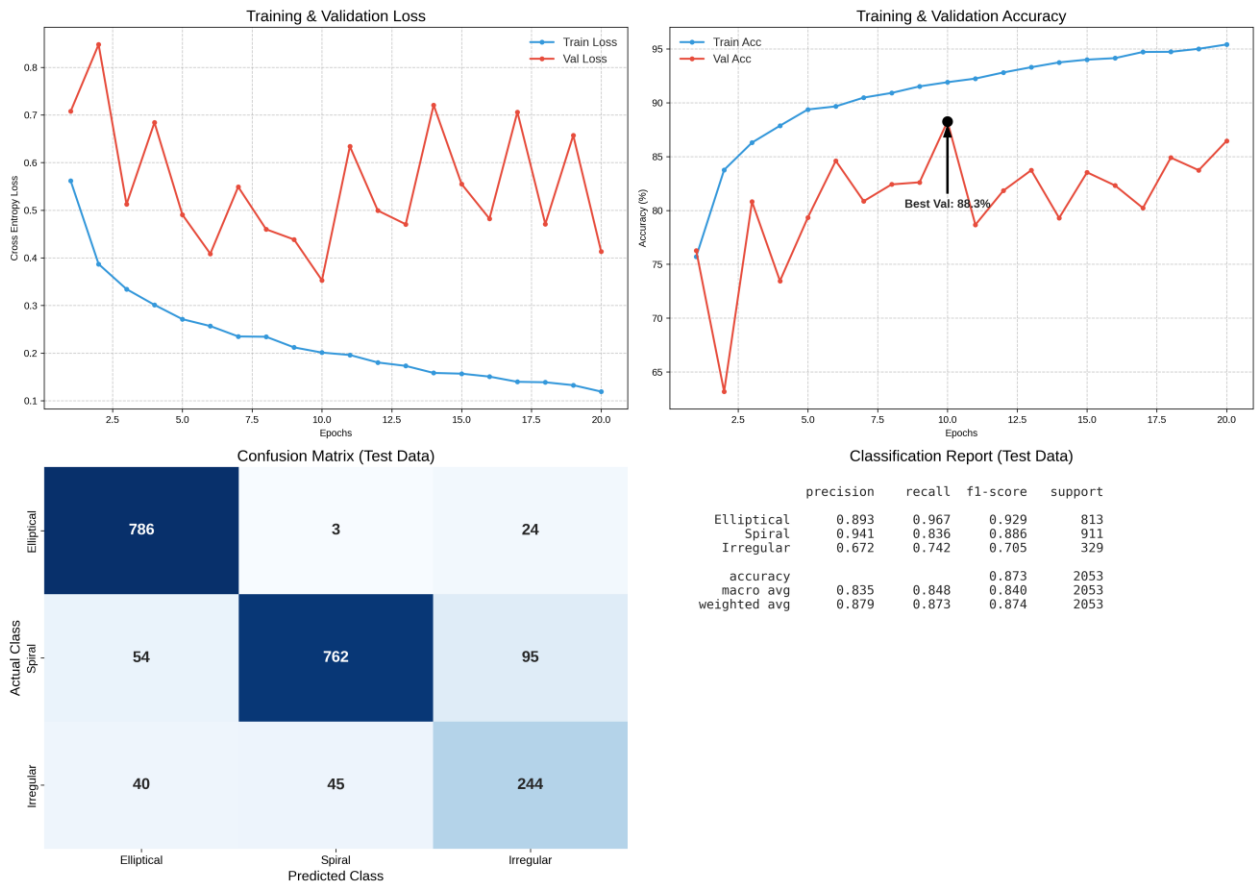
Model Performance Dashboard: RESNET18



ResNet Performance Dashboard – Loss Curve, Accuracy Curve, Confusion Matrix, Metrics

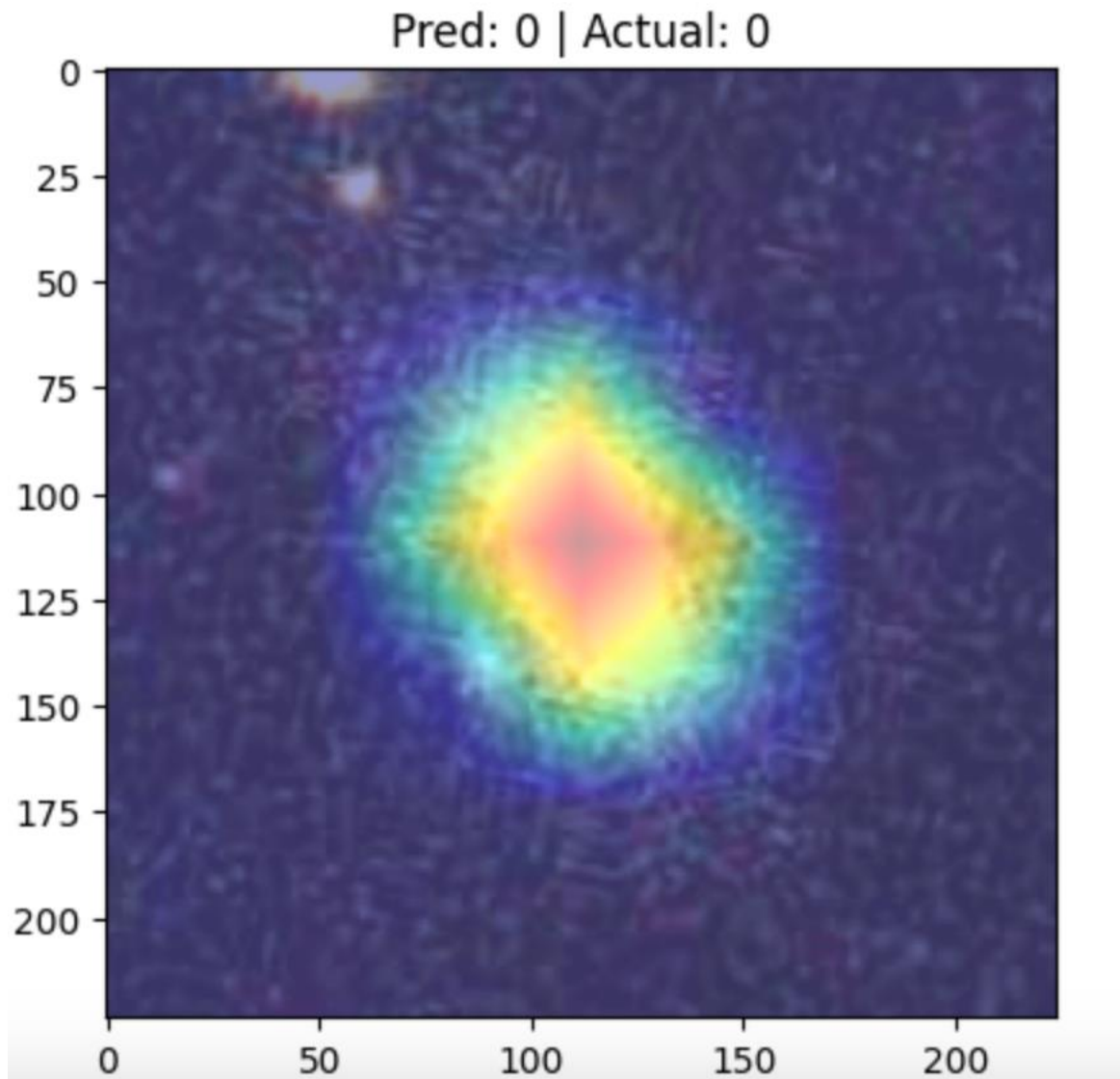
Unfortunately, ResNet performed better than EfficientNet with an accuracy of 92% vs. 87%. This means that the model did not meet the >95% goal. However, it did exceed the 70% goal.

Model Performance Dashboard: EFFICIENTNET_B0



EfficientNet Performance Dashboard – Loss Curve, Accuracy Curve, Confusion Matrix, Metrics

Since there is a strong class imbalance, it is important to look at other metrics besides accuracy. Such as class recall and F1 and macro-avg F1. ResNet has a higher accuracy because it performed better on the majority classes (F1 = 96%, 93%). EfficientNet handled the majority classes worse (F1 = 93%, 89%). But, EfficientNet had a much better recall on the minority class (70% vs 74%), but had worse precision and worse F1 score. This is shown in the confusion matrices where irregular has a higher amount of correct classifications on the EfficientNet model. Regarding the Macro-Average F1, ResNet performed better (88% vs 84%).



Grad-CAM heatmap of elliptical galaxy

This is the output of the Grad-CAM for EfficientNet-b0. The model was able to correctly predict that this is an elliptical galaxy. The heatmap flares up around the galaxy's bright center. The heatmap also was able to identify the galaxy as being featureless and appearing to be a smooth bright blob. Which are all characteristics of an elliptical galaxy. A big win is that the model is not giving any weight to the background stars, one of which is extremely close to the galaxy. This means that the model is able to differential between contaminants and the actual galaxy.

Code Submission

Repository: <https://github.com/TatumSoward/ComputerVisionFinal-GalaxyMorphologyClassifier/>

Project Setup & Execution Guide

1. Environment Requirements

This project is designed to run in a Kaggle Notebook environment using a GPU P104 or T4 x2 accelerator.

Language: Python 3.10+

Framework: PyTorch 2.0+

Dependencies: torchvision, pandas, Pillow, opencv-python (cv2), matplotlib, and tqdm.

2. Dataset & Model Inputs

Before running the cells, ensure the following three inputs are added to your Kaggle sidebar:

Dataset: Galaxy Zoo - The Galaxy Challenge (Contains the raw .jpg images).

Models: GalaxyMorphology-EfficientNetv1 (Contains the trained .pth weights).

Notebook Output: AIG210-GalaxyZooPreprocessing (Contains the train_split.csv, test_split.csv, and norm_stats.json).

3. Execution Steps

Step 1: Data Linking & Unzipping

The code utilizes the /kaggle/temp/ directory for high-speed I/O.

Run the Unzipping cells first. This extracts the image ZIP and the preprocessing output ZIP into the temp folder.

Note: This requires approximately 10GB of temporary disk space.

Step 2: Model Configuration

Set the `eff_net` toggle to True or False depending on which architecture you wish to evaluate.

Ensure the `model_path` string matches the version number in your Kaggle input (e.g., `/v1/` or `/v2/`).

Step 3: Normalization & Transforms

The code automatically loads `norm_stats.json`. This is a critical step; using standard ImageNet stats instead of these custom Galaxy Zoo stats will result in poor Grad-CAM heatmaps.

Step 4: Run the Notebook

The Grad-CAM output will be at the bottom.

References

AstroDave, AstroTom, @ Winton, C. R., Joycenv, & Willett, K. (2013). Galaxy Zoo - The Galaxy Challenge. Retrieved from <https://kaggle.com/competitions/galaxy-zoo-the-galaxy-challenge>

Banerji, M., Lahav, O., Lintott, C., Abdalla, F., Schawinski, K., Bamford, S., Andreescu, D., Murray, P., Raddick, M., Slosar, A., Szalay, A., Thomas, D., & Vandenberg, J. (2010). Galaxy Zoo: reproducing galaxy morphologies via machine learning★: Galaxy Zoo: morphology via machine learning. *Monthly Notices of the Royal Astronomical Society*, 406(1), 342–353.

Nerella, Bhavana & Banala, Smrithi & Konkapaka, Srivani & V, Bhavana. (2026). Galaxy Classification: A Comparative Study. 10.1109/ICAAIC64647.2025.11330315.

Sparke, L. S., & Gallagher, J. S. (2007). *Galaxies in the Universe*. Cambridge University Press.

Willett, K., Lintott, C., Bamford, S., Masters, K., Simmons, B., Casteels, K., Edmondson, E., Fortson, L., Kaviraj, S., Keel, W., Melvin, T., Nichol, R., Raddick, M., Schawinski, K., Simpson, R., Skibba, R., Smith, A., & Thomas, D. (2013). Galaxy Zoo 2: detailed morphological classifications for 304 122 galaxies from the Sloan Digital Sky Survey. *Monthly Notices of the Royal Astronomical Society*, 435(4), 2835–2860.

Submission and Group Work

Submit the completed document by the deadline through Learn@Seneca.

Please include the following declaration at the end of your file:

We, Tatum Soward, Stephane Donald Njike Paho, and Johnpaul Tamburro, declare that the attached project is entirely our own work and has been completed in accordance with the Seneca Academic Policy. We have not copied or reproduced any part of this assignment, either manually or electronically, from any unauthorized source, including but not limited to AI tools, homework-sharing websites, or other students' work, unless explicitly cited as references. We have not shared our work with others, nor have we received unauthorized assistance in completing this project.

Specify what each member has done towards the completion of this project:

	Name	Task(s)
1	Tatum Soward	EDA / Preprocessing, Training ResNet18 (baseline), prepared report, EfficientNet Training (Minor)
2	Johnpaul Tamburro	Grad-CAM, ResNet18 (minor), presentation video
3	Stephane Donald Njike Paho	EfficientNet Training